

PROJECT CHARTER

COURSE

SPM-458

SEMESTER

4th

INSTRUCTOR

Humera Azam

UNIVERSITY OF KARACHI • COMPUTER SCIENCE DEPARTMENT

Software Project Management

ReceiptLedger

Turning a phone photo of any paper receipt into structured, categorized spending data.

6

SPRINTS

OCR

CORE ENGINE

Medium

COMPLEXITY

SUBMITTED BY

#	NAME	ROLL NUMBER	ROLE
001	Aazmeer Sarfaraz Faridy	B24110006002	Member
002	Arish Ahmed Khan	B24110006026	Member
003	Iman Hussain	B24110006054	Member
004	Muhammad Asad Khan	B24110006087	Team Lead
005	Syed Aun Shamsi	B24110006134	Member
006	Zainab Hashmi	B24110006162	Member

Charter prepared for Lab #01 — Project Charter Development, in fulfillment of the Software Project Management-458 lab requirements at the University of Karachi.

01 Project Overview

ReceiptLedger is the team's semester project for Software Project Management-458 (Lab #01: Project Charter Development). It applies pretrained OCR and rule-based categorization to a habit that rarely survives long-term as a spreadsheet exercise — turning a simple phone photo of a receipt into clean, categorized, month-over-month spending data.

PROBLEM STATEMENT

Households and small shop owners keep paper receipts in drawers with no real visibility into where their money goes month to month, because manually entering every receipt into a spreadsheet is tedious enough that almost nobody actually does it.

02 Project Objectives (SMART)

#	OBJECTIVE	SUCCESS MEASURE
O1	Accurate line-item extraction	Reliable OCR accuracy on a sample of real receipts, verified manually
O2	Automatic categorization	Majority of extracted items correctly sorted using rule-based matching
O3	Duplicate detection	Duplicate receipt entries caught to avoid double-counting spend
O4	Clear spending visibility	Simple month-over-month, category-wise spending dashboard

03 Project Scope

IN SCOPE

- ✓ Phone-photo capture of receipts as users naturally collect them.
- ✓ OCR extraction of individual line items and prices, even on faded print.
- ✓ Rule-based categorization of items into spending categories (groceries, fuel, etc.).
- ✓ Duplicate-receipt detection via image or text similarity.
- ✓ Month-over-month, category-wise spending dashboard.

OUT OF SCOPE

- ✗ Bank or card statement reconciliation.
- ✗ Tax filing or invoice-generation features.
- ✗ Categorization beyond common local household/shop keyword coverage.

04 Technology Stack & Tools

The stack leans on pretrained OCR and keyword-rule categorization rather than a trained NLP model, keeping the team's effort on local keyword coverage and duplicate-detection tuning — the parts that matter most for real accuracy.

LAYER	TOOL / APPROACH	PURPOSE
Mobile Capture	Flutter or lightweight web app	Simple photo-upload right after getting a receipt
OCR Engine	Google Cloud Vision API or Tesseract	Extracts printed text from receipts, including faded thermal print
Categorization	Keyword/rule-based matcher (item → category)	Sorts line items into spend categories; ML classifier optional stretch goal
Duplicate Detection	Image hashing or text-similarity check	Flags a receipt photographed twice to avoid double-counting
Backend/Storage	Firebase or PostgreSQL + REST API	Stores categorized line items and spend history
Dashboard	React/Chart.js	Category-wise and monthly spend trend visualization

SYSTEM DATA FLOW



05 Key Deliverables

DELIVERABLE	DESCRIPTION
Capture App	Mobile/web app for photographing receipts as they're collected.
OCR Pipeline	Line-item and price extraction pipeline, tested against a real receipt sample.
Categorization Layer	Rule-based matcher sorting items into local spend categories.
Duplicate Detector	Image/text-similarity check preventing double-counted entries.
Spend Dashboard	Monthly and category-wise spending trend visualization.

06 Stakeholders

STAKEHOLDER	ROLE / INTEREST	EXPECTATION
Households / Small Shop Owners	Primary end users; photograph receipts and view spend trends	An effortless way to actually see where money goes, without spreadsheet habits
Course Instructor (Humera Azam)	Project sponsor and evaluator	A well-scoped, demoable, on-time deliverable per lab requirements
Project Team	Designs, builds, and tests the system	Clear scope, defined roles, and a workable sprint plan
Local Shops (indirect)	Source of receipt formats and naming conventions	Categorization rules that reflect real local keyword patterns

07 Budget & Resource Estimate

CATEGORY	ESTIMATE / APPROACH	
Personnel	6-member student team, semester-duration effort — no monetary cost	
OCR API	Google Cloud Vision API free tier (1,000 units/month) covers pilot testing; Tesseract (open-source, free) as fallback if usage exceeds free tier — est. PKR 0–3,000 for the semester	
Backend/Storage	Firebase Spark (free) plan or Supabase free tier — sufficient for pilot-scale data; est. PKR 0	
Domain / Hosting (Dashboard)	Vercel/Netlify free tier for React dashboard demo; optional custom domain — est. PKR 0–2,500	
Mobile Data / Testing	Data packages for field-testing with household/shop contacts across sprints — est. PKR 1,500	
Printing & Documentation	Charter, report, and presentation printing/binding for submission — est. PKR 1,000	
Contingency	Buffer for paid OCR overage or minor tooling needs — est. PKR 2,000	
Hardware	Personal smartphones and laptops for capture and development — no additional procurement	
Estimated Total Project Cost		PKR 4,500 – 10,000

08 Approval

Humera Azam

Course Instructor, SPM-458

Signature: _____

Date: _____